GRADE 8 PRE-TECHNICAL STUDIES

DETAILED TEACHING NOTES • FULL YEAR • 2027

School: _____ Teacher: _____ TSC No.: _____

Learning Area: Pre-Technical Studies | Grade: 8 | Year: 2027

Curriculum alignment note

These teacher-prepared notes follow the five strands and fourteen sub-strands used in the accompanying Grade 8 Pre-Technical Studies scheme. They provide explanations, examples, practical guidance and safety reminders to support teaching. They are not a reproduction of an official KICD publication. Teachers should cross-check exact curriculum wording, scope and local safety requirements against the current KICD curriculum design and approved learning materials.

General safety reminder

Practical activities involving fire, electrical equipment, cutting tools, chemicals, kilns or workshop machinery must be risk-assessed, teacher-supervised and conducted only with suitable equipment and protective measures. Learners should never be asked to handle live electrical wiring, operate a kiln or fight an uncontrolled fire.

1.0 FOUNDATIONS OF PRE-TECHNICAL STUDIES

1.1 Fire Safety

Meaning and importance of fire safety

Fire safety refers to precautions and practices used to prevent unwanted fires, reduce the risk of injury and protect property and the environment. Fire can be useful in cooking, heating and industrial processes, but an uncontrolled fire can cause burns, smoke inhalation, destruction of property and loss of life.

Causes of fire outbreaks

Common causes include unattended open flames, careless handling of matches and candles, faulty electrical wiring, overloaded sockets, cooking accidents, leaking gas, smoking materials and the unsafe storage of flammable liquids. A fire requires heat, fuel and oxygen; removing one of these can help stop combustion.

Flammable and combustible materials

Flammable materials ignite easily, while combustible materials can burn when sufficient heat is supplied. Examples may include petrol, paraffin, alcohol-based products, paper, dry vegetation, cloth and some plastics. Store such materials safely, away from heat, sparks and naked flames.

Prevention of fire outbreaks

Keep exits clear; switch off appliances when not in use; avoid overloading electrical sockets; replace damaged cables; keep matches away from children; store fuels in suitable labelled containers; maintain cooking equipment; and follow workplace safety instructions.

Fire classes and extinguishing methods

Different fires involve different fuels. Water may be suitable for some ordinary solid-material fires but must never be used on live electrical equipment or burning oil. Appropriate extinguishers are selected according to the fire type and the label/instructions. Learners should not attempt to fight a fire that is spreading or beyond their training.

Basic firefighting principles

Cooling removes heat; smothering limits oxygen; starving removes or isolates fuel; and interrupting the chemical chain reaction stops combustion. A trained person may use an appropriate extinguisher while keeping a clear escape route. Raise an alarm, evacuate and call emergency services when necessary.

Safe response to fire

Alert others, activate the alarm if available, leave by a safe route, do not use lifts during a building fire, stay low if smoke is present, and assemble at the designated safe point. Do not re-enter a burning building. Follow the school or workplace emergency plan.

1.2 Data Safety

Data and information

Data are raw facts such as numbers, words, images or measurements. Information is data that has been processed or organised to make it meaningful. Electronic devices store and process data for communication, learning, record keeping and decision-making.

Importance of data

Data may include school work, contacts, photographs, financial records and project files. Losing or exposing data can interrupt work, cause embarrassment or lead to financial and privacy harm. Data should therefore be handled responsibly.

Threats to data

Threats include malware (malicious software), viruses, phishing messages, weak passwords, unauthorised access, accidental deletion, device theft, hardware failure and unsafe downloads. Learners should avoid opening suspicious links or attachments and should seek adult or technical support when uncertain.

Protecting data

Use strong, unique passwords or passphrases; keep software updated; use reputable security tools; lock devices; restrict access; avoid sharing passwords; download only from trusted sources; and sign out of shared devices. Do not enter personal information on unfamiliar websites.

Backups and recovery

A backup is an additional copy of important data kept separately from the original. Backups may be stored on approved external media or secure cloud services. Back up important work regularly and check that copies can be opened. Recovery procedures depend on the device and service.

Responsible digital behaviour

Respect other people's privacy and intellectual property. Ask permission before sharing photographs or personal information. Do not access another person's files or accounts without permission. Report cyberbullying, suspicious messages or possible data breaches to a trusted adult.

2.0 COMMUNICATION IN PRE-TECHNICAL STUDIES

2.1 Plane Geometry

Drawing instruments and their uses

Common instruments include drawing board, T-square or parallel rule, set squares, ruler, compass, divider, protractor, pencils and eraser. Each instrument supports accurate measurement, line work, angles, circles and geometric construction.

Drawing environment

A good drawing environment is clean, well-lit, ventilated and organised. Keep instruments in good condition, secure drawing paper properly and avoid clutter. Correct posture and careful handling improve accuracy and reduce strain.

Geometric constructions

Plane geometry deals with two-dimensional shapes and their properties. Learners use instruments to construct and combine lines, angles, triangles, quadrilaterals, circles and other plane figures. Accurate construction depends on correct measurement, neat line work and checking dimensions.

Applications

Geometric constructions are used in technical drawings, patterns, product design, building layouts, craft work and planning. Learners should explain how accuracy in geometry supports communication of design ideas.

2.2 Dimensioning

Meaning and purpose

Dimensioning is the process of showing the sizes and positions of features on a technical drawing. It communicates the measurements needed to make, inspect or assemble an object.

Types of dimensions

Linear dimensions show lengths, widths or heights. Radial dimensions indicate the radius of an arc or circle, while diameter dimensions indicate the full distance across a circle through its centre. Angular dimensions show the size of an angle; arc dimensions may indicate the length of a curved portion.

Dimensioning lines

Extension lines indicate the points between which a measurement applies. Dimension lines carry the measurement and usually have arrowheads or another accepted terminator. Leader lines connect a note or dimension to a feature. Centre lines identify centres and axes.

Methods of dimensioning

Parallel dimensioning places dimensions from a common reference line. Chain dimensioning places dimensions in sequence from one feature to the next. Combined dimensioning uses both methods where suitable. Follow the conventions demonstrated by the teacher or approved drawing reference.

Good practice

Use clear, legible figures; avoid unnecessary repetition; place dimensions where they can be read easily; do not allow dimension lines to cross where possible; and use consistent units and conventions. Learners should practise dimensioning simple shapes before complex drawings.

2.3 Plain Scale Drawing

Meaning and purpose

A plain scale is a graduated line used to represent actual lengths in a drawing at a stated scale. It helps a user read or set measurements that are larger or smaller than the real object.

Scale and representation

A scale compares drawing length with actual length. For example, a scale of 1:100 means one unit on the drawing represents 100 of the same units in reality. Always use the same units before calculating or reading a scaled measurement.

Reading a scale

Identify the scale statement, locate the zero mark and read the graduations carefully. Check whether the drawing uses millimetres, centimetres or metres. Estimate only where the scale and task permit it.

Drawing to scale

To draw a figure to scale, convert each actual measurement using the scale ratio, mark the scaled lengths accurately and construct the figure with appropriate instruments. Label the scale and dimensions clearly.

Applications

Plain scales are useful in technical drawing, maps, floor plans, site layouts and product design. A scale drawing communicates proportions while allowing a large object or area to fit on a page.

2.4 Visual Programming

Meaning and applications

Visual programming uses graphical blocks, icons or visual elements to create instructions for a computer. It can help beginners understand program logic without typing every command in a text-based language.

Programming applications

Examples used for learning include Scratch and Microsoft MakeCode. Applications may support animations, games, interactive stories, simulations and simple control projects. Availability depends on the device and school resources.

Input, processing and output

Input is information entered into a system, processing is the work performed according to instructions, and output is the result. In a visual program, input may be a key press or mouse click, processing may be a sequence of blocks, and output may be movement, sound or a displayed message.

Blocks and instructions

Visual programming environments provide blocks for events, movement, appearance, sound, control, sensing and variables. Blocks are arranged in a meaningful sequence to tell the computer what to do.

Sequence, events and variables

Sequence is the order in which instructions run. An event starts an action, such as clicking a flag or pressing a key. A variable stores a value that may change while the program runs.

Testing and debugging

Run the program and compare its behaviour with the intended result. A bug is an error in a program; debugging is the process of locating and correcting it. Test one change at a time and keep a working copy.

Creating a simple project

Plan the purpose, identify inputs and outputs, arrange instructions, test the project, correct errors and explain how it works. Save work responsibly and respect other people's projects and digital content.

3.0 MATERIALS FOR PRODUCTION

3.1 Composite Materials

Meaning and examples

A composite material is made by combining two or more distinct materials so that the resulting material has useful properties. The components remain identifiable at a material level, even though the finished product behaves as one material.

Concrete

Concrete commonly contains cement, sand, gravel or crushed stone and water. Cement paste binds the aggregates. Concrete is widely used in foundations, floors, roads and structural elements. Reinforcement may be added where tensile strength is required.

Bricks and manufactured boards

Bricks are commonly made from clay or other suitable earth materials shaped and fired. Manufactured boards such as plywood and particleboard are made by bonding layers, particles or fibres with adhesives under controlled conditions.

Paper-mâché and coated materials

Paper-mâché combines paper with paste or adhesive to form moulded objects. Plastic-coated paper combines paper with a coating that changes its resistance to moisture or improves its surface properties.

Properties and selection

Composite materials are selected according to strength, weight, durability, cost, resistance to moisture or heat, appearance and intended use. Their properties depend on the components and manufacturing process.

Responsible use

Choose materials suited to the task, reduce unnecessary waste, reuse or recycle where practical, and follow safety instructions when cutting, mixing, sanding or handling composite materials.

3.2 Ceramics

Meaning and characteristics

Ceramics are inorganic, non-metallic materials commonly produced by shaping and heating mineral-based raw materials. Many ceramics are hard, durable and heat-resistant, but may be brittle and can crack when struck or subjected to sudden temperature changes.

Raw materials

Clay is a common ceramic raw material. Other ingredients may include silica and feldspar, depending on the product. The mixture and preparation affect workability, shrinkage, strength and final appearance.

Shaping and drying

Ceramic products may be shaped by hand-building, moulding, throwing on a wheel or pressing. The shaped object is dried carefully to remove moisture and reduce cracking during firing.

Firing and glazing

Firing heats the dried product in a kiln to develop its ceramic properties. Some products are glazed to create a smooth, decorative or less absorbent surface, then may be fired again according to the process.

Properties and uses

Ceramics are used for pottery, tiles, bricks, sanitary ware, tableware, electrical insulators and some heat-resistant components. Selection depends on strength, porosity, heat resistance and intended service.

Care and safety

Handle ceramic items carefully because edges and broken pieces can cut. Use protective equipment and follow teacher supervision when working with clay, glazes, kilns or tools. Kiln operation is for trained adults.

4.0 TOOLS AND PRODUCTION

4.1 Cutting Tools

Purpose and examples

Cutting tools remove or separate material to obtain a required shape or size. Examples include snips, chisels, handsaws, hacksaws, planes, scrapers, knives, wire strippers and cutters. The correct tool depends on the material and task.

Selection

Consider the material, thickness, shape required, cutting direction, tool condition and safety requirements. Do not use a tool for a purpose for which it was not designed.

Safe handling

Inspect tools before use. Keep hands away from cutting edges, secure the workpiece, use the correct protective equipment and cut in a direction that does not put the body in the path of the tool. Work under supervision and follow demonstrations.

Maintenance

Clean tools after use, dry them where necessary, protect metal surfaces from corrosion, sharpen only with appropriate guidance and report damaged tools. Store cutting tools with edges protected and out of reach of unauthorised users.

Practical work

A practical task should begin with identifying the material and tool, planning the cut, securing the workpiece, demonstrating the technique, completing the task safely and inspecting the result. Learners should explain why the selected tool was appropriate.

4.2 Computer Software

Meaning and categories

Computer software is a set of instructions that tells computer hardware what to do. System software manages the computer and provides a platform for other programs. Application software helps users perform specific tasks.

System software

Operating systems manage files, memory, devices and user interaction. Utility programs support maintenance and protection. Device drivers help the operating system communicate with hardware.

Application software

Examples include word processors, spreadsheets, presentation programs, drawing/design applications, browsers and communication tools. Each supports different tasks in education, business and production.

Using software

Select an application suited to the task, open or create a file, enter or manipulate information, save the work with a meaningful filename and close the application properly. Respect licensing and avoid installing unknown software.

Importance in the workplace

Software supports communication, design, record keeping, calculations, presentations, research and planning. Users should choose appropriate tools and protect information stored or shared through them.

5.0 ENTREPRENEURSHIP

5.1 Bookkeeping

Meaning and importance

Bookkeeping is the systematic recording of financial transactions of a business. Accurate records help a business owner know what was received, what was paid, what is owed and how the business is performing.

Transactions and source documents

A transaction is an event involving money or value. Source documents provide evidence of transactions. Examples include receipts, invoices, payment vouchers and bank records. Keep documents organised and record transactions promptly.

Assets, liabilities and capital

Assets are resources owned or controlled by a business. Liabilities are amounts the business owes. Capital is the owner's investment in the business. These terms help describe the financial position of a business.

Recording transactions

Record the date, description, amount and relevant account in a consistent manner. Distinguish money received from money paid. Check entries against source documents and correct errors transparently.

Cash records

A simple cash book records cash received and cash paid. A petty cash record tracks small routine payments. The balance should be checked against the cash available and supporting documents.

Accuracy and ethics

Bookkeeping requires honesty, confidentiality, completeness and accuracy. Do not falsify records or conceal transactions. Records should be stored securely and made available only to authorised people.

Using records

Business records help owners plan purchases, monitor expenses, set prices, identify debts and make decisions. Learners can practise with a simple imaginary enterprise and sample transactions.

5.2 Income and Budgeting

Income

Income is money or value received by an individual or household. Sources may include wages, salaries, business profit, farming, casual work, gifts or other lawful sources. Income may be regular or irregular.

Budgeting

A budget is a plan showing expected income and expenditure over a stated period. It helps people plan spending, meet needs, save and avoid unnecessary debt.

Needs and wants

Needs are essentials such as food, shelter, clothing, education and healthcare. Wants are things people would like to have but can often postpone. A budget should prioritise needs and consider available resources.

Preparing a simple budget

List expected income, estimate essential expenses, identify other planned spending and savings, then compare total expenses with income. If expenses exceed income, revise the plan by reducing or postponing non-essential spending.

Ethical financial practices

Use income honestly, avoid fraud and misleading claims, respect other people's money and keep accurate records. Do not borrow without understanding repayment obligations.

Reviewing a budget

Compare actual income and expenditure with the plan. Explain differences and adjust future estimates. A budget is a guide that can be revised when circumstances change.

5.3 Marketing Goods and Services

Meaning and purpose

Marketing involves identifying customer needs and communicating how goods or services may meet them. It helps customers learn about products and helps businesses reach potential buyers.

Goods and services

Goods are tangible items, while services are activities or benefits provided to customers. Examples in the community can be used to identify different customers and their needs.

Marketing methods

Methods include personal selling, posters, signs, demonstrations, word of mouth, radio, print and digital platforms. The method should suit the product, audience, cost and available resources.

Product, price, place and promotion

These are common marketing considerations: what is offered, what it costs, where it is available and how customers learn about it. Learners can use these ideas to plan a simple marketing message.

Ethical marketing

Marketing should be truthful, fair and respectful. Avoid false claims, hidden charges, misuse of customer information and discriminatory messages. Give accurate information about product features and limitations.

Creating a marketing message

Identify the intended customer, describe the product or service accurately, state its useful features and provide clear contact or access information where appropriate. Review the message for honesty and clarity.

5.4 Distribution of Goods and Services

Meaning of distribution

Distribution is the process of making goods or services available to consumers. It involves movement, storage, handling and delivery from producers or providers to users.

Channels of distribution

A direct channel moves goods from producer to consumer without an intermediary. An indirect channel uses one or more intermediaries such as wholesalers and retailers.

Role of intermediaries

Wholesalers may buy in bulk and sell to retailers. Retailers sell goods to final consumers. Transporters, agents and other service providers may support movement and access to markets.

Choosing a channel

The channel may depend on the type of product, distance, cost, storage needs, customer location and speed of delivery. Perishable goods may require careful handling and timely delivery.

Ethics in distribution

Fair distribution involves honest weights and measures, safe handling, clear pricing, reliable delivery and respect for customers. Avoid hoarding, deliberate shortages, counterfeits and misleading practices.

Community importance

Effective distribution links producers with consumers, creates employment, supports trade and helps communities access goods and services. Learners can map a familiar product's route from producer to consumer.

REVISION AND APPLICATION QUESTIONS

1. Explain the fire triangle and describe four practical ways of preventing fire outbreaks.
2. Distinguish between data and information. Identify four threats to data and explain how each may be reduced.
3. Name five drawing instruments and state one use for each.
4. Explain the difference between linear, radial, angular and arc dimensioning.
5. A drawing uses a scale of 1:100. Explain what this means and describe how a real measurement is represented.
6. Describe the steps involved in planning, testing and debugging a simple visual program.
7. Define a composite material and give three examples, stating a use for each.
8. Outline the main stages in producing a ceramic item from clay.
9. Select suitable cutting tools for three different materials and justify each selection. State the safety precautions required.
10. Distinguish between system software and application software, giving examples.
11. Explain why source documents are important in bookkeeping. Give four examples.
12. Prepare a simple weekly budget using an imaginary income and explain how you prioritised needs.
13. Develop an ethical marketing message for a product or service available in your community.
14. Draw a simple distribution channel for a locally available product and explain the role of each intermediary.

Suggested learner-centred approaches

Use demonstrations, guided practical work, group investigations, observation of local examples, drawing and design tasks, role-play, digital exploration, project work, peer feedback and reflective discussion. Adapt tasks to available resources while maintaining the intended learning outcomes and safety requirements.